Recurrent neural networks

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class 5

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Section 1

Intro

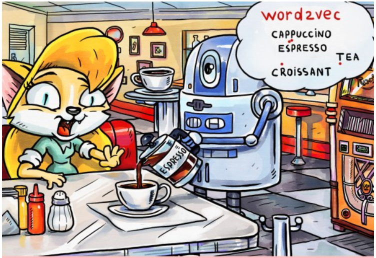
Last weeks

In the last 2 weeks we learned about vector embeddings and fully-connected artificial neural networks (both are trained by some version of gradient descent):

Concepts and ideas:

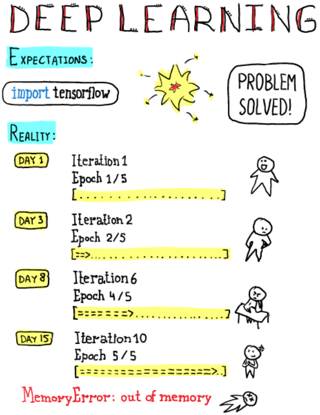
1. Vector embeddings are unsupervised models for natural language processing. They assign vectors to words in a way to capture semantic relationships, e.g.,

$$v(\text{China}) + v(\text{capital}) = v(\text{Beijing})$$



- Espresso? But I ordered a cappuccino!
- Don't worry, the cosine distance between them is so small that they are almost the same thing.

2. Neural networks are complex supervised models. They require a careful choice of hyperparameters and training settings, including regularization.



Learning outcomes of today's class

1. Train a recurrent neural network (RNN). In particular, know what input data for an RNN looks like (sequential data).
2. Explain architecture of many-to-one, one-to-many, many-to-many RNN. We will do it in week 6.
3. Explain advantages and disadvantages of an LSTM (long short-term memory) layer over a plain vanilla RNN layer. We will explain LSTM in detail in week 6.

Revision of neural networks — WooClap

- https://app.wooclap.com/ZGDCZZ

Figure 1: Scan this QR code to open WooClap activity

Section 2

Recurrent neural networks

Bag of words

GloVe produces feature vectors for individual words. To get a feature vector for the whole text, we just add all feature vectors (or add an normalize) for individual words together. Some meaning is lost in the process.

Example

Compare:

- MFE is better than MBA
- MBA is better than MFE
- better is MFE MBA than

Applications of RNN

Recurrent neural networks (RNN) are used to process sequential data, like sequence of word vectors, sequence of daily adjusted close prices of a number of stock exchange indices etc.

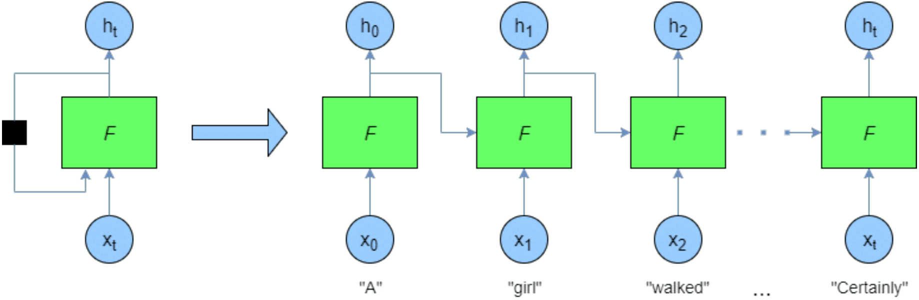
Some applications:

- Next word prediction.
- Text classification, such as semtiment analysis.
- Time series anomaly detection
- Stock market prediction
- Text summarization
- Document generation
- Automating business processes
- Chatbots
- Music composition.

Basic RNN

Source of image:

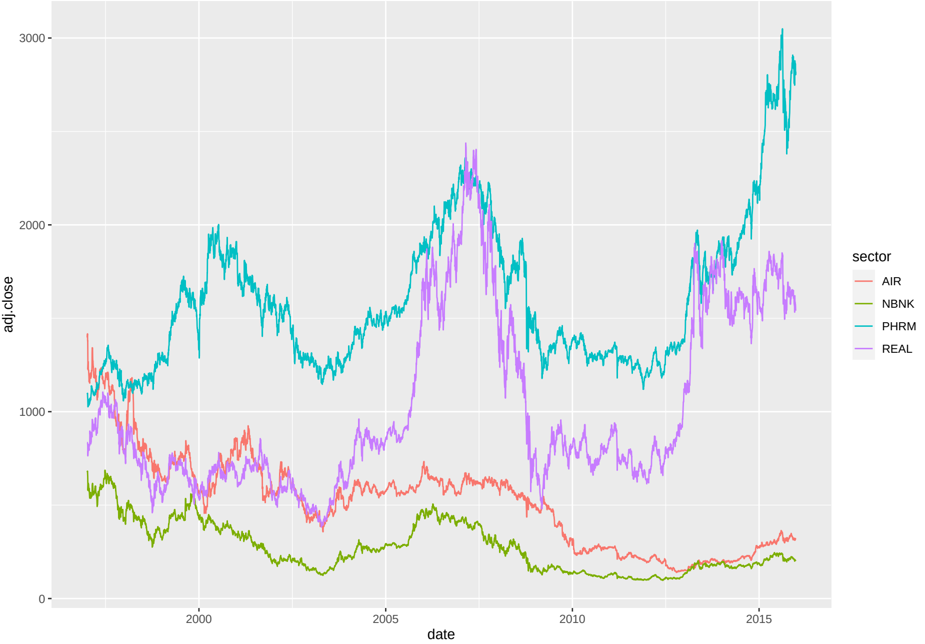
- https://adventuresinmachinelearning.com/recurrent-neural-networks-lstm-tutorial-tensorflow/



Example: Nikkei 225 data

Trading data on sectors comprising Nikkei 225

date	NBNK	AIR	REAL	PHRM
2015-07-31	240.29	353.66	1763.98	2845.55
2006-04-21	485.62	640.44	1703.74	2100.47
2000-07-12	368.94	751.74	781.96	1967.20
2011-06-09	105.19	219.61	716.99	1278.43
1997-07-01	657.90	1185.34	1061.07	1255.40
2008-05-14	293.77	592.55	1535.60	1759.85
2010-05-27	130.34	241.58	759.34	1244.35


3000
2000
1000
0
adj.close
2000
2005
2010
2015
date
sector
AIR
NBNK
PHRM
REAL

Let's say that we want to predict PHRM based on 5 days of historical prices of NBNK, AIR, REAL

date	NBNK	AIR	REAL	PHRM
1997-05-19	607.6	1234.47	1061.93	1222.9
1997-05-20	595.9	1235.09	1059.69	1219.9
1997-05-21	583.7	1221.42	1035.93	1198.6
1997-05-22	590.6	1229.43	1031.53	1203.5
1997-05-23	597.8	1230.39	1052.71	1218.3

Let us say that U is

	1	2	3
1	1.7	0.6	0.6
2	1.7	0.1	0.8
3	-0.9	0.9	-0.2
4	1.3	-1.5	0.9

And W is

	1	2	3	4
1	1.7	1.9	1.6	-1.7
2	-1.0	-1.5	-1.4	0.1
3	-0.2	-0.1	2.0	-0.4
4	1.8	0.2	1.8	1.6

And the bias vector b is

	1
1	-0.2
2	1.3
3	1.0
4	1.2

Here is our data:

date	NBNK	AIR	REAL	PHRM
1997-05-19	607.6	1234.47	1061.93	1222.9
1997-05-20	595.9	1235.09	1059.69	1219.9
1997-05-21	583.7	1221.42	1035.93	1198.6
1997-05-22	590.6	1229.43	1031.53	1203.5
1997-05-23	597.8	1230.39	1052.71	1218.3

So x_1 is

	1
1	607.60
2	1234.47
3	1061.93

And let's say that $h_0 = 0$ and that the activation function is linear rectifier. Then $z_1 = Ux_1 + Ws_0 + b$, which is

	1
1	2410.76
2	2005.91
3	351.80
4	-106.09

After applying the activation, we get $h_1 = f_{rnn}(z_1)$, which is

	1
1	2410.76
2	2005.91
3	351.80
4	0.00

Here is our data:

date	NBNK	AIR	REAL	PHRM
1997-05-19	607.6	1234.47	1061.93	1222.9
1997-05-20	595.9	1235.09	1059.69	1219.9
1997-05-21	583.7	1221.42	1035.93	1198.6
1997-05-22	590.6	1229.43	1031.53	1203.5
1997-05-23	597.8	1230.39	1052.71	1218.3

So x_2 is

	1
1	595.90
2	1235.09
3	1059.69

Now $z_2 = Ux_2 + Wh_1 + b$ is

	1
1	10862.10
2	-3926.55
3	385.18
4	5250.74

Applying the activation, we get $h_2 = f_{rnn}(z_2)$, which is

	1
1	10862.10
2	0.00
3	385.18
4	5250.74

Here is our data:

date	NBNK	AIR	REAL	PHRM
1997-05-19	607.6	1234.47	1061.93	1222.9
1997-05-20	595.9	1235.09	1059.69	1219.9
1997-05-21	583.7	1221.42	1035.93	1198.6
1997-05-22	590.6	1229.43	1031.53	1203.5
1997-05-23	597.8	1230.39	1052.71	1218.3

So x_3 is

	1
1	583.70
2	1221.42
3	1035.93

Now $z_3 = Ux_3 + Wh_2 + b$ is

	1
1	12502.10
2	-8931.80
3	-3134.59
4	28506.51

Applying the activation, we get $h_3 = f_{rnn}(z_3)$, which is

	1
1	12502.10
2	0.00
3	0.00
4	28506.51

Here is our data:

date	NBNK	AIR	REAL	PHRM
1997-05-19	607.6	1234.47	1061.93	1222.9
1997-05-20	595.9	1235.09	1059.69	1219.9
1997-05-21	583.7	1221.42	1035.93	1198.6
1997-05-22	590.6	1229.43	1031.53	1203.5
1997-05-23	597.8	1230.39	1052.71	1218.3

So x_4 is

	1
1	590.60
2	1229.43
3	1031.53

Now $z_4 = Ux_4 + Wh_3 + b$ is

	1
1	-24847.10
2	-7697.96
3	-13533.38
4	67967.40

Applying the activation, we get $h_4 = f_{rnn}(z_4)$, which is

	1
1	0.0
2	0.0
3	0.0
4	67967.4

Here is our data:

date	NBNK	AIR	REAL	PHRM
1997-05-19	607.6	1234.47	1061.93	1222.9
1997-05-20	595.9	1235.09	1059.69	1219.9
1997-05-21	583.7	1221.42	1035.93	1198.6
1997-05-22	590.6	1229.43	1031.53	1203.5
1997-05-23	597.8	1230.39	1052.71	1218.3

So x_5 is

	1
1	597.80
2	1230.39
3	1052.71

Now $z_5 = Ux_5 + Wh_4 + b$ is

	1
1	-113158.66
2	8779.51
3	-26827.17
4	108628.03

Applying the activation, we get $h_5 = f_{rnn}(z_5)$, which is

	1
1	0.00
2	8779.51
3	0.00
4	108628.03

And now the vector h_5 is passed to the output layer where we multiply it by, say, the matrix

	1	2	3	4
1	3.2	0.35	-5.2	-0.02

And add the bias (in this case it is a scalar, but it may be a vector for, say, multiclass classification)

	1
1	260.9

Then we get $z_{out} = w_{out} \cdot h_5 + b_{out}$, which is

	1
1	1161.17

Section 3

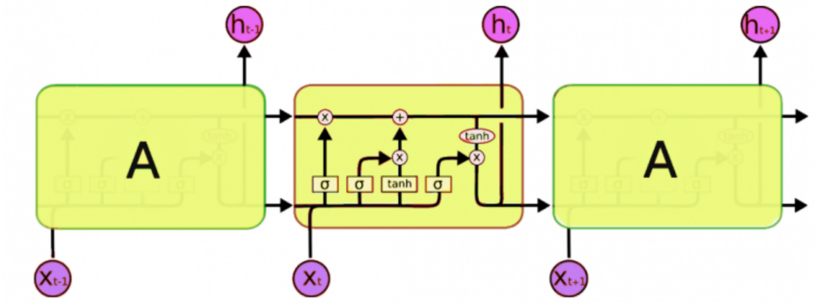
LSTM

LSTM

Vanishing gradient problem

Source of image:

- https://www.analyticsvidhya.com/blog/2017/12/fundamentals-of-deep-learning-introduction-to-lstm/



Team assignment 2 (graded): Notebook 9

Approx 90 minutes

Section 4

Summary

Main takeaways

Recurrent neural networks process sequential data, such as time series, texts, speech etc. Basic RNN architecture is theoretically capable of having some memory to do that, but in practice it doesn't work very well because of vanishing gradients. Usually, in practice we use LSTM (long short-term memory) units to build RNN.

Word embeddings can be integrated with supervised machine learning models for prediction. In particular, pretrained word embeddings such as GloVe can be integrated into recurrent neural networks.

Recurrent neural networks are prone to overfitting. Regularization can be done by adjusting the dropout rates. There are two kinds of dropout rates — dropout for output of an RNN layer and dropout inside an RNN layer.

Concepts and ideas

1. Sequential data
2. Recurrent neural network.
3. Vanishing gradients
4. LSTM

At home

Notebook 10: recurrent neural network integrated with pre-trained word embeddings.

CLICK ME

There are no more theoretical problems. You can now focus on your project work.